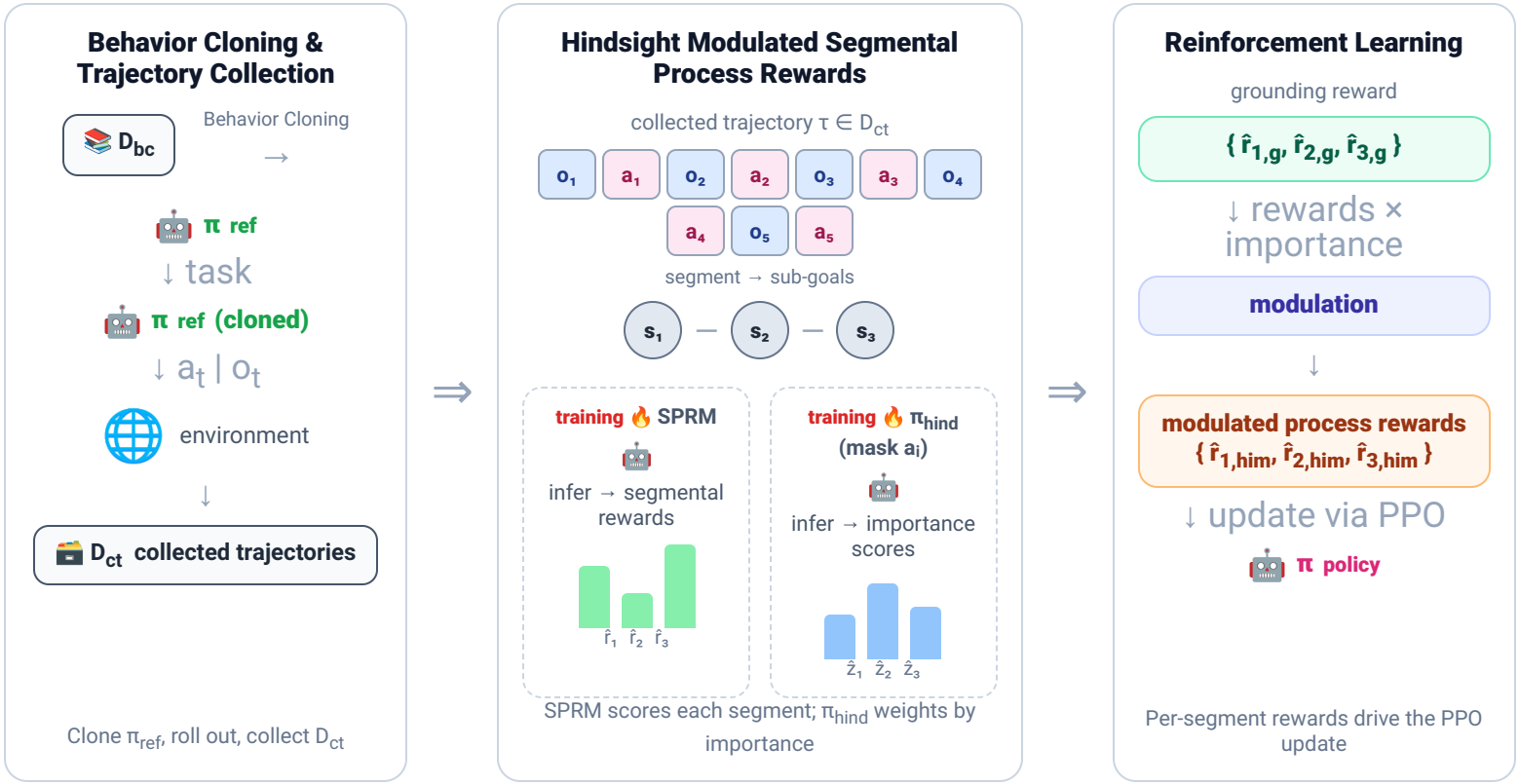


SeRPO: Segment-level Rubric Process Rewards for AppWorld Agents



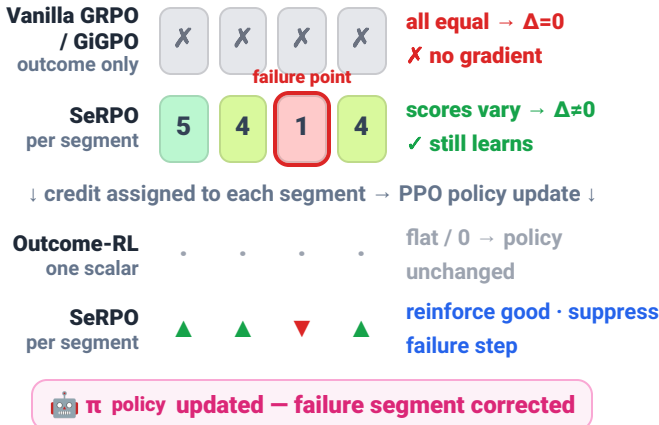
Why segment-level process rewards matter

LEARNS WHERE OUTCOME RL CAN'T

On **fail-only** trajectories every rollout shares the same outcome, so outcome-reward baselines (Vanilla GRPO, GiGPO) get equal rewards $\rightarrow$ group variance 0 $\rightarrow$ **gradient = 0** (no learning). SeRPO's **per-segment** rewards still vary within a trajectory $\rightarrow$ **gradient $\neq$ 0**, so the policy keeps improving even when all trajectories failed.

LOCATES THE FAILURE POINT

A single outcome scalar says only *pass / fail*. Per-segment rewards $\{\hat{r}_1, \hat{r}_2, \hat{r}_3\}$ attribute credit along the trajectory, so a low-scoring segment **pinpoints which step caused the failure** (e.g. a premature action with missing API exploration) — turning an opaque rollout into a diagnosable, interpretable signal.



Same failed trajectory. Outcome-RL spreads one scalar (0 here) over every token $\rightarrow$ no targeted push. SeRPO sends **+credit** to the segments that worked and **-credit** to the failure step, so PPO updates the policy *precisely where it went wrong*.